

Class 12

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Section 1: Proportion of G/G in a population

Downloaded a CSV file from Ensembl https://useast.ensembl.org/Homo_sapiens/Variation/Sample?db=core;r=17:39895045-39895145;v=rs8067378;vdb=variation;vf=959672880#373531_tablePanel

Q5: What proportion of the Mexican Ancestry in Los Angeles sample population (MXL) are homozygous for the asthma associated SNP (G|G)? [HINT: You can filter the displayed genotypes by entering the population code MXL. Then either count those of interest or download a CVS file for this population and use excel or the R functions read.csv(), and table() to answer this question]

Here we read this CSV file

```
mxl <-read.csv("373531-SampleGenotypes-Homo_sapiens_Variation_Sample_rs8067378.csv")
head (mxl)
```

	Sample..	Male..	Female..	Unknown..	Genotype..	forward..	strand..	Population..	s..	Father
1					NA19648	(F)		A A	ALL, AMR, MXL	-
2					NA19649	(M)		G G	ALL, AMR, MXL	-
3					NA19651	(F)		A A	ALL, AMR, MXL	-
4					NA19652	(M)		G G	ALL, AMR, MXL	-
5					NA19654	(F)		G G	ALL, AMR, MXL	-
6					NA19655	(M)		A G	ALL, AMR, MXL	-
	Mother									
1										-

```
2      -
3      -
4      -
5      -
6      -
```

```
table(mx1$Genotype..forward.strand.)
```

```
A|A  A|G  G|A  G|G
22  21  12   9
```

```
table(mx1$Genotype..forward.strand.) / nrow(mx1) * 100
```

```
      A|A      A|G      G|A      G|G
34.3750 32.8125 18.7500 14.0625
```

Q6. Back on the ENSEMBLE page, use the “search for a sample” field above to find the particular sample HG00109. This is a male from the GBR population group. What is the genotype for this sample?

Now let’s look at a different population. I picked the GBR.

```
gbr <- read.csv("373522-SampleGenotypes-Homo_sapiens_Variation_Sample_rs8067378.csv")
```

Find proportion of GIG

```
round(table(gbr$Genotype..forward.strand.) / nrow(gbr) * 100, 2)
```

```
      A|A      A|G      G|A      G|G
25.27 18.68 26.37 29.67
```

This variant that is associated with childhood asthma is more frequent in the GBR population than the BKL population.

Let’s now dig into this further.

Section 4: Population Scale Analysis [OPTIONAL]

One sample is obviously not enough to know what is happening in a population. You are interested in assessing genetic differences on a population scale. So, you processed about ~230 samples and did the normalization on a genome level. Now, you want to find whether there is any association of the 4 asthma-associated SNPs (rs8067378...) on ORMDL3 expression.

How many samples do we have?

```
expr <- read.table("genotype.txt")
head(expr)
```

	sample	geno	exp
1	HG00367	A/G	28.96038
2	NA20768	A/G	20.24449
3	HG00361	A/A	31.32628
4	HG00135	A/A	34.11169
5	NA18870	G/G	18.25141
6	NA11993	A/A	32.89721

```
nrow(expr)
```

```
[1] 462
```

```
table(expr$geno)
```

```
A/A A/G G/G
108 233 121
```

```
library(ggplot2)
```

Let's make a box plot

```
ggplot(expr) +
  aes(geno, exp, fill=geno) +
  geom_boxplot(notch=TRUE)
```

